

18.06 Linear Algebra — Lecture 10 & 11
Textbook Notes — Sections 3.5

◦ Section 3.5

• Architecture

Four Fundamental Subspaces → complete picture of matrix A .
↳ Come in pairs living in two different spaces.

In $\mathbb{R}^n$:

• Row space $C(A^T)$ — dimension (r)

• Nullspace $N(A)$ — dimension ($n-r$)

In $\mathbb{R}^m$:

• Column space $C(A)$ — dimension (r)

• Left nullspace $N(A^T)$ — dimension ($m-r$)

↳ Dimensions always add up correctly: $r + (n-r) = n$
 $r + (m-r) = m$

• The Four Subspaces

1. Row Space $C(A^T)$ — all linear combinations of the rows of A .
— Equivalently the column space of A^T
— Basis: nonzero rows of R .

2. Column Space $C(A)$ — all vectors Ax
— Basis: the pivot columns of A (not R)

3. Nullspace $N(A)$ — all solutions to $A^T y = 0$, equivalently $y^T A = 0$
— ~~The y vector multiplies from the left → Left Nullspace~~
— ~~Basis: found by applying elimination to A^T~~
— Basis: the $n-r$ special solutions

4. Left Nullspace $N(A^T)$ — all solutions to $A^T y = 0$, equivalently $y^T A = 0$
— y vector multiplies from the left → "left nullspace"
— Basis: found by applying elimination to A^T

• What does / does not Change Under Elimination

- Row space & nullspace $\rightarrow$ same for A & R (row operations)
- Column space & Left nullspace $\rightarrow$ different for A & R
 $\hookrightarrow$ dimensions remain the same.
- $\rightarrow$ Always take Basis from $C(A)$ not $C(R)$.

• Rank One Matrices

Any rank one matrix can be written as $A = \underline{u} \underline{v}^T$ (one col u & one row v)

- Column Space: spanned by $\underline{u}$ - a single line.
- Row Space: spanned by $\underline{v}$ - a single line
- Nullspace: everything perpendicular to $\underline{v}$, dimension $n-1$.
- Left Nullspace: everything perpendicular to $\underline{u}$, dimension $m-1$.

• Example: Incidence Matrix.

Strang introduces graph with 4 nodes & 5 edges.

Incidence matrix A has one $+1$ and one -1 in each row.

One for the node an edge leaves, one for the node it enters.

$\rightarrow$ The four subspaces have direct physical meaning:

- Nullspace of A : all nodes at the same potential - $x = c(1,1,1,1)$
- Left nullspace of A : combinations of edges forming loops -
 $\hookrightarrow$ Kirchhoff's voltage law

• Row space: differences of potentials - no loops.

• Column space: solvability condition on currents - Kirchhoff's Current Law

$\hookrightarrow$ Four subspaces encode real structure. Rank = 3, confirming $r = n-1$ for a connected graph.

• Key Theorem

For an $m \times n$ matrix of rank r :

$$\dim C(A^T) + \dim N(A) = r + (n - r) = \underline{n}$$

$$\dim C(A) + \dim N(A^T) = r + (m - r) = \underline{m}$$

↳ Row space and column space always have the same dimension r , even though they live in different spaces ($\mathbb{R}^n$ and $\mathbb{R}^m$).

- 18.06 Linear Algebra - Lecture 12
Textbook Notes - Section 10.1

Section 10.1 - Graphs & Networks.

Overview

Section 10.1 applies the four fundamental subspaces directly to electrical networks. (abstract subspaces "become" physical laws).

A graph has nodes and edges.

The matrix which captures this structure is the incidence matrix, A .

The Incidence Matrix

For a graph with m edges and n nodes, A is $m \times n$.

Each row corresponds to one edge

Entry $+1$ in the column of the node the edge leaves.

Entry -1 in the column of the node the edge enters.

All other entries zero.

↳ Every row has exactly one $+1$ and one -1

The Four Subspaces & Their Physical Meaning

• Nullspace $N(A) \rightarrow$ solutions to $Ax = 0$ where x is a potential at each node.

$\rightarrow Ax = 0$ means every edge has zero potential diff., so all nodes have the same potential.

$\rightarrow$ Special Sol $(1, 1, 1, 1)$ confirms: dimension 1,
rank $= n - 1$ for a
connected graph.

• Row Space $C(A^T) \rightarrow$ dimension $r = n - 1$
 $\rightarrow$ These are potential differences with no loops
 $\hookrightarrow$ a spanning tree

• Column Space $C(A) \rightarrow$ dimension r
 $\rightarrow$ Solvability condition $Ax = b$ requires b currents to satisfy Kirchhoff's ~~voltage~~^{current} law: ~~the sum~~
 $\hookrightarrow$ total current into each node equals total current out
 $\rightarrow b$ must be in the column space.

• Left Nullspace $N(A^T) \rightarrow$ solutions to $A^T y = 0$.
 $\rightarrow$ Combinations of edges that form loops.
 $\rightarrow$ Kirchhoff's voltage law: sum of voltage drops around any loop is zero
 $\rightarrow$ Dimension $m - r = m - n + 1$
 $\hookrightarrow$ equals number of independent loops.

• Ohm's Law & Kirchhoff's Laws as Linear Algebra:

The full network problem connects three equations:

• $Ax = b \rightarrow$ potential differences across edges (Kirchhoff's voltage)

• $y = CAx \rightarrow$ currents from potentials via Ohm's Law
 (C is a diagonal conductance matrix)

• $A^T y = f \rightarrow$ current balance at nodes (Kirchhoff's current)

Combined $\Rightarrow$ $A^T C A x = f$

$\hookrightarrow$ fundamental equation of network analysis, and it appears identically in structural mechanics, fluid flow, and resistor networks

• Euler Formula

For a connected graph: nodes - edges + loops = 1

equiv: $n - m + (m - n + 1) = 1$

$\hookrightarrow$ Dimension count of four subspaces restated geometrically